

TECHBIG SOLUTIONS

Ambattur, Chennai

SAP APO – ECC E2E Complete Training Syllabus

Advanced Planning and Optimization | SAP ECC / SCM APO

Program Objective: Build practical consultant-level capability in SAP Advanced Planning and Optimization (APO) integrated with SAP ECC, covering Demand Planning (DP), Supply Network Planning (SNP), Production Planning & Detailed Scheduling (PP/DS), Global Available-to-Promise (GATP), CIF integration, configuration, testing and production support.

Reference basis: SAP Help documentation for SAP APO 7.0 EHP3 and SAP APO/ECC integration, including DP, SNP, PP/DS, GATP and CIF.

Program	SAP APO – ECC E2E
Level	Functional Consultant / Implementation / Support
Core Areas	DP SNP PP/DS GATP CIF ECC Integration
Delivery Focus	Configuration + Real-time scenarios + Production support
Suggested Duration	80–120 hours (customizable by batch and project depth)

1. SAP APO & Supply Chain Management Overview

- SAP APO purpose and architecture
- APO vs SAP ECC roles
- Strategic, tactical and operational planning
- APO components: DP, SNP, PP/DS and Global ATP
- APO deployment with SAP SCM / ECC integration
- End-to-end supply-chain planning flow

2. SAP APO Technical Architecture

- APO/SCM Server architecture
- ECC OLTP and APO system landscape
- LiveCache fundamentals
- Application, database and interface layers
- Development, quality and production landscapes
- Background jobs and technical monitoring

3. SAP APO Organizational & Master Data Concepts

- Planning area, planning version and model
- Location, product and resource master data
- Transportation lanes and procurement relationships
- Production process models (PPM)
- Quota arrangements and sourcing
- Master-data consistency and governance

4. Core Interface (CIF) – ECC to APO

- CIF architecture and integration models
- Initial data transfer and change transfer
- Material, customer, vendor and location integration
- BOM, routing/work center and production master integration
- Stock, sales orders, purchase orders and planned orders
- CIF queues, errors, reprocessing and reconciliation

5. APO Demand Planning (DP) – Fundamentals

- Demand planning concepts and process
- Planning objects and characteristics
- Planning areas, key figures and time buckets
- Planning books and data views
- Interactive planning and macros
- Historical demand and forecast preparation

6. APO Demand Planning – Forecasting

- Statistical forecasting concepts
- Forecast models and parameters
- Moving average, exponential smoothing and regression
- Forecast error and accuracy measures
- Consensus demand planning
- Promotion, lifecycle and causal factors
- Forecast release to supply planning

7. APO Demand Planning – Configuration & Operations

- DP master-data setup
- Planning area and key-figure configuration
- Forecast profiles and forecast execution
- Background jobs and planning processes
- Planning book design
- DP troubleshooting and data consistency

8. APO Supply Network Planning (SNP) – Fundamentals

- SNP objectives and planning horizons
- Network model and supply chain structure
- Locations, products and transportation lanes
- Demand/supply balancing
- Stock and safety-stock concepts
- Aggregated and detailed planning

9. SNP Planning Methods

- Heuristics-based planning

- CTM – Capable-to-Match overview
- SNP optimizer concepts
- Rule-based planning
- Deployment and replenishment planning
- Quota and sourcing logic
- Pegging and supply-demand relationships

10. SNP Configuration & Execution

- SNP planning area and key figures
- Planning books and interactive planning
- SNP heuristics profiles
- Optimizer profiles and constraints
- Deployment profiles
- Supply planning alerts and exception handling

11. APO Production Planning & Detailed Scheduling (PP/DS)

- PP/DS architecture and purpose
- Product and resource planning
- Production process models
- PPM/PDS concepts
- Planning procedures and planning file concepts
- Procurement proposals and production orders

12. PP/DS Detailed Scheduling

- Detailed scheduling board
- Finite capacity planning
- Sequence-dependent setup times
- Setup matrices
- Heuristics and optimization
- Pegging and order relationships
- Resource overload and exception resolution

13. PP/DS Integration with ECC PP

- ECC production master data to APO
- Planned orders and production orders
- Order transfer between ECC and APO
- Confirmations and goods movements
- Execution in ECC and planning in APO
- Consistency checks and reconciliation

14. APO Global Available-to-Promise (GATP)

- ATP architecture and business scenarios
- Check instructions and scope of check

- Product availability checks
- Rules-based ATP
- Product allocation
- Backorder processing concepts
- Integration with sales order processing

15. GATP Advanced Functions

- Multilevel ATP
- Capable-to-Promise (CTP)
- ATP with PP/DS
- ATP simulation and explanation tools
- Substitution and alternative locations
- Delivery date determination
- Exception and confirmation analysis

16. APO ATP Configuration & Integration

- ECC-SD and APO ATP integration
- Integration models and CIF
- Check control and business rules
- ATP master data
- Sales-order confirmation flow
- Troubleshooting ATP inconsistencies

17. APO Integration with ECC MM

- Material and purchasing master data
- Purchase requisitions and purchase orders
- External procurement proposals
- Stock transport orders
- Vendor and purchasing integration
- CIF transfer and execution flow

18. APO Integration with ECC SD

- Sales orders and demand transfer
- Deliveries and ATP
- Customer/material master integration
- Demand planning from sales history
- Global ATP confirmation flow
- Order and delivery reconciliation

19. APO Integration with ECC PP

- BOM, routing and work-center integration
- Production versions / PPM concepts
- Planned and production order integration

- Capacity and resource integration
- Production execution feedback
- PP/DS-ECC reconciliation

20. APO Integration with ECC MM-IM / Inventory

- Stock transfer and inventory visibility
- Stock categories and availability
- Goods receipt/goods issue impacts
- Inventory consistency
- Planning-relevant stock
- Physical inventory considerations

21. APO Transportation Planning Overview

- Transportation planning concepts
- Transportation lanes and transportation resources
- TLB – Transport Load Builder overview
- Vehicle scheduling concepts
- Integration with ECC logistics execution
- When transportation functions are used in APO

22. APO Alerts & Exception Management

- Alert Monitor
- Exception messages
- Planning alerts
- ATP exceptions
- Resource overloads and delays
- Alert profiles and notification concepts
- Root-cause analysis

23. APO Reporting & Monitoring

- Product view and product planning table
- Planning books and analytics
- Plan Monitor
- CIF monitoring
- Application logs and background jobs
- Data consistency and reconciliation reports

24. APO Configuration – IMG & Core Settings

- APO basic settings
- Model/version setup
- Planning area and key-figure configuration
- DP/SNP/PPDS/GATP configuration principles
- CIF integration configuration

- Number ranges and background jobs
- Transport strategy

25. APO Performance & Housekeeping

- LiveCache monitoring concepts
- Planning performance
- Background job scheduling
- Data volume management
- Reorganization and housekeeping
- Performance troubleshooting
- System consistency checks

26. APO Security & Authorizations

- APO user administration
- Roles and authorization concepts
- Planning-book and transaction access
- RFC/integration security
- Technical users
- Audit and production access controls

27. APO Implementation Methodology

- Requirement gathering and blueprinting
- Supply-chain process mapping
- Master-data readiness
- Configuration and development
- Integration testing
- User acceptance testing
- Cutover and go-live
- Hypercare and stabilization

28. APO Production Support & Troubleshooting

- CIF queue errors
- Master-data mismatches
- Planning run failures
- PP/DS scheduling issues
- ATP confirmation issues
- Performance and job failures
- Data reconciliation between ECC and APO
- Incident and root-cause analysis

29. Real-Time E2E Project Scenarios

- Demand forecast → SNP supply plan → PP/DS production plan
- ECC sales order → GATP availability confirmation

- Forecast → SNP deployment → ECC procurement
- ECC PP master data → PP/DS scheduling → ECC execution
- CIF initial load and change transfer recovery
- Capacity bottleneck and finite scheduling project
- Multi-location replenishment and stock balancing
- ATP shortage with alternative-location strategy
- Planning exception and alert-management project
- End-to-end APO production support project

30. Interview & Practical Assessment

- APO consultant interview questions
- DP/SNP/PPDS/GATP scenario questions
- CIF troubleshooting questions
- ECC integration scenarios
- Configuration case studies
- Production-support ticket simulations
- End-to-end implementation walkthrough
- Resume/project explanation and mock interview

Recommended Hands-on Deliverables

- Configure and execute an APO-ECC CIF integration model with initial and change transfer.
- Build a DP planning book and execute statistical forecasting with forecast release.
- Create an SNP supply plan using heuristics and analyze shortages/alerts.
- Create and schedule PP/DS orders using finite capacity and detailed scheduling.
- Configure and execute an ECC sales-order Global ATP scenario.
- Perform end-to-end reconciliation from ECC master/transaction data to APO planning results and back to ECC execution.

Training Outcome

After completing this syllabus, learners should be able to understand APO architecture, configure and operate core APO planning processes, integrate APO with SAP ECC using CIF, analyze planning and ATP exceptions, execute end-to-end supply-chain scenarios, and support APO in implementation and production environments.

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